

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
In [2]: df = pd.read_csv("Train.csv")
df
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/ 2117
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 1759
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O. 310128
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	11380
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	37345
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	21153
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	11205
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./ 660
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	11136
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	37037

891 rows × 12 columns

In [4]: df.head()

```
Out[4]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450

```
In [5]: df.columns
```

```
Out[5]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
              'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
              dtype='object')
```

```
In [29]: df.dtypes
```

```
Out[29]: PassengerId    int64
Survived              int64
Pclass               int64
Name                 object
Sex                 int64
Age                 object
SibSp               int64
Parch               int64
Ticket              object
Fare               float64
Cabin              object
Embarked           object
dtype: object
```

```
In [6]: df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	3
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	4
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	5
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	6
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	7
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	8
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	5

In [7]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null    int64
1   Survived        891 non-null    int64
2   Pclass          891 non-null    int64
3   Name            891 non-null    object
4   Sex             891 non-null    object
5   Age             714 non-null    float64
6   SibSp           891 non-null    int64
7   Parch           891 non-null    int64
8   Ticket          891 non-null    object
9   Fare            891 non-null    float64
10  Cabin           204 non-null    object
11  Embarked        889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [10]: `df.shape`

Out[10]: (891, 12)

In [11]: `df.isnull().sum()`

```
Out[11]: PassengerId     0
Survived               0
Pclass                 0
Name                   0
Sex                    0
Age                   177
SibSp                  0
Parch                  0
Ticket                 0
Fare                   0
Cabin                 687
Embarked               2
dtype: int64
```

```
In [12]: def fun1(value):  
        if value == "male":  
            return 1  
        elif value == "female":  
            return 0  
        else:  
            return None
```

```
In [15]: df["Sex"].value_counts()
```

```
Out[15]: Sex  
1      577  
0      314  
Name: count, dtype: int64
```

```
In [13]: df['Sex'] = df['Sex'].apply(fun1)
```

```
In [14]: df
```

Out[14]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	1	27.0	0	0	211536
887	888	1	1	Graham, Miss. Margaret Edith	0	19.0	0	0	112053
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	0	NaN	1	2	W./C. 6607
889	890	1	1	Behr, Mr. Karl Howell	1	26.0	0	0	111369
890	891	0	3	Dooley, Mr. Patrick	1	32.0	0	0	370376

891 rows × 12 columns

```
In [18]: def fun2(value):
         if (np.isnan(value) == False):
             return int(value)
```

```
In [19]: df["Age"] = df["Age"].apply(fun2)
```

```
In [20]: df.head()
```

Out[20]:	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	PC 17599	71
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8

```
In [21]: df_age_greater_than_25 = df[df["Age"] > 25.0]
```

```
In [22]: df_age_greater_than_25.head()
```

Out[22]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	PC 17599	71
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8
6	7	0	1	McCarthy, Mr. Timothy J	1	54.0	0	0	17463	51

In [24]: `df["Age"].value_counts()`

Out[24]:

Age	
24.0	31
22.0	27
28.0	27
30.0	27
18.0	26
..	
66.0	1
53.0	1
80.0	1
12.0	1
74.0	1

Name: count, Length: 71, dtype: int64

In [25]:

```
def fun3(value):
    if value < 20:
        return "teenager"
    elif 20 <= value < 40:
        return "young"
    elif 40 <= value < 60:
        return "middle aged"
    elif value >= 60:
        return "senior citizen"
    else:
        return None
```

In [27]: `df["Age"] = df["Age"].apply(fun3)`

In [28]: `df.head(10)`

Out[28]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	1	young	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	young	1	0	PC 17596
2	3	1	3	Heikkinen, Miss. Laina	0	young	0	0	STON/O2 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	young	1	0	113803
4	5	0	3	Allen, Mr. William Henry	1	young	0	0	373450
5	6	0	3	Moran, Mr. James	1	None	0	0	330871
6	7	0	1	McCarthy, Mr. Timothy J	1	middle aged	0	0	17463
7	8	0	3	Palsson, Master. Gosta Leonard	1	teenager	3	1	349909
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	0	young	0	2	347742
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	0	teenager	1	0	237736

In []: